

Project Report — AI-Powered Smart Recruitment Assistant

1. Objective

Recruiters running training/enrollment programs need to know which candidates are actually likely to be **looking for a job change**, so they can focus recruitment effort where it will pay off instead of contacting the entire candidate pool blindly. This project builds and compares three classification models that predict this outcome (`target` : 1 = looking for a job change, 0 = not) from a candidate's profile: education, experience, current employer, and training activity.

2. Data overview

- **Source:** `aug_train.csv` (labeled) + `aug_test.csv` (unlabeled), combined for consistent cleaning then split back apart.
- **Combined size:** 21,287 rows.
- **Labeled rows used for modeling:** 19,158.
- **Class balance:** 4,777 of 19,158 labeled candidates (**24.93%**) are looking for a job change — a meaningfully imbalanced target, roughly 1:3.
- **Features used:** `city_development_index` , `relevent_experience` , `enrolled_university` , `education_level` , `major_discipline` , `experience` , `company_size` , `company_type` , `last_new_job` , `training_hours` .
- **Dropped:** `enrollee_id` (identifier), `city` (high-cardinality, redundant with `city_development_index`), `gender` (largely missing).

Missing data was present in 7 of 10 features prior to cleaning, most heavily in `company_size` (6,560 missing) and `company_type` (6,774 missing) — consistent with many candidates not currently being employed or not disclosing employer details.

3. Modeling approach

Three baseline classifiers were trained on an 80/20 stratified train/test split (15,326 / 3,832 rows) of the labeled data, using standardized features:

Model	Key settings
Logistic Regression	<code>max_iter=1000</code> , <code>class_weight='balanced'</code>
Decision Tree	<code>criterion='gini'</code> , <code>max_depth=7</code> , <code>class_weight='balanced'</code>
Gaussian Naive Bayes	default settings

`class_weight='balanced'` was used across models to counteract the 25/75 class imbalance, prioritizing the ability to catch job-seeking candidates (recall on class 1) rather than optimizing plain accuracy.

4. Results (held-out test set, n = 3,832)

Model	Accuracy	Precision	Recall	F1 Score	AUC
Decision Tree	0.7737	0.5323	0.7602	0.6261	0.7934
Naive Bayes	0.7516	0.5015	0.5236	0.5123	0.7532
Logistic Regression	0.7145	0.4531	0.7037	0.5513	0.7684

Decision Tree is the best-performing model on this dataset, ranking first on F1 score, AUC, recall, and accuracy — it catches 76% of candidates who are actually looking for a job change while keeping the false-positive rate manageable.

Per-class detail — Decision Tree (best model)

	Precision	Recall	F1-score	Support
Not looking (0)	0.91	0.78	0.84	2,877
Looking (1)	0.53	0.76	0.63	955

Per-class detail — Logistic Regression

	Precision	Recall	F1-score	Support
Not looking (0)	0.88	0.72	0.79	2,877
Looking (1)	0.45	0.70	0.55	955

Per-class detail — Naive Bayes

	Precision	Recall	F1-score	Support
Not looking (0)	0.84	0.83	0.83	2,877
Looking (1)	0.50	0.52	0.51	955

Interpretation: all three models trade off precision for recall on the minority class (job-seekers), which is the intended effect of `class_weight='balanced'` — the practical cost of missing an actual job-seeking candidate is treated as worse than the cost of a wasted outreach to someone who wasn't actually looking. Naive Bayes is the most conservative (least improvement in recall, but somewhat better precision than the other two); Decision Tree gives the best overall trade-off.

5. What drives the prediction (feature importance, Decision Tree)

Feature	Importance
<code>city_development_index</code>	0.473
<code>company_size</code>	0.322
<code>major_discipline</code>	0.072
<code>experience</code>	0.029
<code>training_hours</code>	0.028
<code>last_new_job</code>	0.026
<code>company_type</code>	0.015
<code>enrolled_university</code>	0.015
<code>education_level</code>	0.011
<code>relevent_experience</code>	0.010

Two features — `city_development_index` and `company_size` — account for roughly **80%** of the Decision Tree's decision-making. Candidates from less-developed cities and those at smaller (or unknown-size) companies are disproportionately more likely to be job-seeking. Everything else (education, major, experience, training hours) plays a comparatively minor role.

6. Business insights (raw, human-readable rates)

Looking at the original (unencoded) labels, the overall rate of candidates looking for a job change is **24.93%** (4,777 of 19,158). Breaking this down:

By education level: | Education | Job-change rate | |---|---| | Graduate | 27.98% | | Masters | 21.44% | | High School | 19.53% | | Phd | 14.01% | | Primary School | 13.31% |

Graduates are the most likely to be job-seeking; PhDs and those with only primary education are the least likely — plausibly reflecting career stage and specialization/stability effects.

By current company type: | Company type | Job-change rate | |---|---| | Other | 23.97% | | Early Stage Startup | 23.55% | | Public Sector | 21.99% | | NGO | 18.62% | | Pvt Ltd | 18.08% | | Funded Startup | 13.99% |

Candidates at funded startups show the lowest job-change rate; those at early-stage startups and public sector roles are more likely to be looking elsewhere.

By relevant experience: | Relevant experience | Job-change rate | |---|---| | No relevant experience | 33.84% | | Has relevant experience | 21.47% |

Candidates **without** relevant experience are ~1.6x more likely to be job-seeking than those with it — likely candidates trying to break into the field.

By enrollment status: | Enrollment | Job-change rate | |---|---| | Full time course | 38.09% | | Part time course | 25.21% | | No enrollment | 21.14% |

Candidates currently enrolled full-time in a course are the strongest signal of active job searching — nearly double the rate of those not enrolled at all.

7. Recommendations

1. **Prioritize outreach** using the Decision Tree model's predictions, focusing first on candidates flagged as likely job-seekers with low `city_development_index` and small/unknown `company_size` — these are the strongest signals in the data.
2. **Segment campaigns:** candidates enrolled full-time in courses and those without relevant experience represent the highest-propensity groups for a job change and may respond best to active recruitment.
3. **Improve the feature set** before deploying further: reintroduce `city` (encoded properly, e.g., target or frequency encoding) instead of dropping it, and switch `company_type` / `enrolled_university` / `major_discipline` from `LabelEncoder` to one-hot encoding to remove artificial ordinality.
4. **Tune the Decision Tree further** (e.g., grid search over `max_depth`, `min_samples_leaf`) or try ensemble methods (Random Forest, Gradient Boosting) — these typically outperform a single tree on tabular data like this and would likely raise the ceiling above the current 0.626 F1 score.
5. **Fix the leakage risk:** refit missing-value imputation and label encoders on the training split only, not the full train+test frame, before any production use.

8. Summary

Of the three baseline models tested, the **Decision Tree (max_depth=7, class-balanced)** is the clear winner (F1 = 0.626, AUC = 0.793, Recall = 0.760 on the job-seeking class), and it points to **city development level** and **company size** as the two dominant predictors of whether a trained candidate is actively looking for a new job. This gives a solid, explainable starting point for a "smart recruitment assistant" — with clear next steps (better encoding, more models, hyperparameter tuning) to improve it further.